

Partial Fractions

ANSWER KEY



Distinct linear factors

- Express $(5x + 7)/((x + 1)(x + 2))$ in partial fractions.
Write $(5x + 7)/((x + 1)(x + 2)) = A/(x + 1) + B/(x + 2)$. Multiply through: $5x + 7 = A(x + 2) + B(x + 1)$.
At $x = -1$: $2 = A(1)$, so $A = 2$. At $x = -2$: $-3 = B(-1)$, so $B = 3$. Result: $2/(x + 1) + 3/(x + 2)$.
- Express $(3x - 1)/(x(x - 3))$ in partial fractions.
Write $(3x - 1)/(x(x - 3)) = A/x + B/(x - 3)$. Then $3x - 1 = A(x - 3) + Bx$. At $x = 0$: $-1 = -3A$, so $A = 1/3$. At $x = 3$: $8 = 3B$, so $B = 8/3$. Result: $1/(3x) + 8/(3(x - 3))$.

Repeated linear factors

- Express $(4x + 1)/(x - 1)^2$ in partial fractions of the form $A/(x - 1) + B/(x - 1)^2$.
 $4x + 1 = A(x - 1) + B$. At $x = 1$: $5 = B$. Comparing coefficients of x : $A = 4$. Result:
 $4/(x - 1) + 5/(x - 1)^2$.
- Express $(2x^2 - 3x + 1)/(x(x - 1)^2)$ in partial fractions of the form $A/x + B/(x - 1) + C/(x - 1)^2$.
 $2x^2 - 3x + 1 = A(x - 1)^2 + Bx(x - 1) + Cx$. At $x = 0$: $1 = A(1)$, so $A = 1$. At $x = 1$: $0 = C(1)$, so $C = 0$.
Comparing x^2 coefficients: $2 = A + B$, so $B = 2 - 1 = 1$. Result: $1/x + 1/(x - 1)$.

Irreducible quadratic factors

- Express $(3x^2 + 2)/((x - 1)(x^2 + 1))$ in partial fractions of the form $A/(x - 1) + (Bx + C)/(x^2 + 1)$.
 $3x^2 + 2 = A(x^2 + 1) + (Bx + C)(x - 1)$. At $x = 1$: $5 = 2A$, so $A = 5/2$. Comparing x^2 coefficients:
 $3 = A + B$, so $B = 3 - 5/2 = 1/2$. Comparing constants: $2 = A - C$, so $C = A - 2 = 1/2$. Result:
 $5/(2(x - 1)) + (x + 1)/(2(x^2 + 1))$.

Applying partial fractions to integration

- Use partial fractions to evaluate $\int (5x + 7)/((x + 1)(x + 2)) dx$, using the decomposition found earlier in this worksheet.
From earlier: $(5x + 7)/((x + 1)(x + 2)) = 2/(x + 1) + 3/(x + 2)$. So
 $\int (5x + 7)/((x + 1)(x + 2)) dx = 2\ln|x + 1| + 3\ln|x + 2| + C$.

Applying partial fractions to a binomial series expansion

- 7 Express $1/((1-x)(1-2x))$ in partial fractions, then use the result to write the first three terms of its series expansion in ascending powers of x .
- $1/((1-x)(1-2x)) = A/(1-x) + B/(1-2x)$. Then $1 = A(1-2x) + B(1-x)$. At $x = 1$: $1 = A(-1)$, so $A = -1$. At $x = 1/2$: $1 = B(1/2)$, so $B = 2$. Result: $-1/(1-x) + 2/(1-2x)$. Expanding each geometric series:
- $$-1(1+x+x^2+\dots) + 2(1+2x+4x^2+\dots) = (-1+2) + (-1+4)x + (-1+8)x^2 + \dots = 1 + 3x + 7x^2 + \dots$$

Combining terms before decomposing

- 8 Express $(7-x)/((x+1)(x-2))$ in partial fractions.
- Write $(7-x)/((x+1)(x-2)) = A/(x+1) + B/(x-2)$. Then $7-x = A(x-2) + B(x+1)$. At $x = -1$: $8 = A(-3)$, so $A = -8/3$. At $x = 2$: $5 = B(3)$, so $B = 5/3$. Result: $-8/(3(x+1)) + 5/(3(x-2))$.
- 9 Express $(x^2+x+1)/(x(x+1))$ in partial fractions. (Hint: the degree of the numerator equals the degree of the denominator, so first perform polynomial division.)
- Since numerator and denominator are both degree 2, divide first:
- $$(x^2+x+1)/(x^2+x) = 1 + 1/(x^2+x) = 1 + 1/(x(x+1)).$$
- Decompose $1/(x(x+1)) = A/x + B/(x+1)$: $1 = A(x+1) + Bx$. At $x = 0$: $A = 1$. At $x = -1$: $B = -1$. Result: $1 + 1/x - 1/(x+1)$.

Applying partial fractions to a rate problem

- 10 A population model gives the rate of change $dP/dt = 1/(P(10-P))$. Express $1/(P(10-P))$ in partial fractions of the form $A/P + B/(10-P)$.
- $1 = A(10-P) + BP$. At $P = 0$: $1 = 10A$, so $A = 1/10$. At $P = 10$: $1 = 10B$, so $B = 1/10$. Result: $1/(10P) + 1/(10(10-P))$ — a decomposition commonly used to solve logistic-growth differential equations by separation of variables.

Partial fractions with three distinct linear factors

- 11 Express $(6x^2-2x-1)/((x-1)(x+1)(x-2))$ in partial fractions.
- Write $(6x^2-2x-1)/((x-1)(x+1)(x-2)) = A/(x-1) + B/(x+1) + C/(x-2)$. Then
- $$6x^2-2x-1 = A(x+1)(x-2) + B(x-1)(x-2) + C(x-1)(x+1).$$
- At $x = 1$: $3 = A(2)(-1) = -2A$, so $A = -3/2$. At $x = -1$: $7 = B(-2)(-3) = 6B$, so $B = 7/6$. At $x = 2$: $19 = C(1)(3) = 3C$, so $C = 19/3$.

Partial fractions with an improper fraction and repeated factor

- 12** Express $x^3/(x-1)^2$ in partial fractions, first performing polynomial division since the numerator has higher degree than the denominator.
 Divide x^3 by $(x-1)^2 = x^2 - 2x + 1$: $x^3 \div (x^2 - 2x + 1) = x + 2$ remainder $3x - 2$. So
 $x^3/(x-1)^2 = x + 2 + (3x - 2)/(x-1)^2$. Decompose the remainder:
 $(3x - 2)/(x-1)^2 = A/(x-1) + B/(x-1)^2$, giving $3x - 2 = A(x-1) + B$. At $x = 1$: $1 = B$. Comparing x coefficients: $A = 3$. Result: $x + 2 + 3/(x-1) + 1/(x-1)^2$.

Using partial fractions to sum a series

- 13** Express $1/(n(n+1))$ in partial fractions, and use the result to find
 $1/(1 \cdot 2) + 1/(2 \cdot 3) + 1/(3 \cdot 4) + 1/(4 \cdot 5)$ by telescoping.
 $1/(n(n+1)) = 1/n - 1/(n+1)$ (standard decomposition with $A = 1$, $B = -1$). Summing from $n = 1$ to 4 :
 $(1 - 1/2) + (1/2 - 1/3) + (1/3 - 1/4) + (1/4 - 1/5) = 1 - 1/5 = 4/5$, since all the middle terms cancel (telescope).

Decomposing a fraction with a repeated quadratic-adjacent structure

- 14** Express $(x+5)/((x-1)(x+1)^2)$ in partial fractions of the form $A/(x-1) + B/(x+1) + C/(x+1)^2$.
 $x+5 = A(x+1)^2 + B(x-1)(x+1) + C(x-1)$. At $x = 1$: $6 = A(4)$, so $A = 3/2$. At $x = -1$: $4 = C(-2)$, so $C = -2$. Comparing x^2 coefficients: $0 = A + B$, so $B = -3/2$. Result:
 $3/(2(x-1)) - 3/(2(x+1)) - 2/(x+1)^2$.
- 15** Express $(2x^2 + x - 1)/(x^2(x+1))$ in partial fractions of the form $A/x + B/(x^2) + C/(x+1)$.
 $2x^2 + x - 1 = Ax(x+1) + B(x+1) + Cx^2$. At $x = 0$: $-1 = B(1)$, so $B = -1$. At $x = -1$: $0 = C(1)$, so $C = 0$. Comparing x^2 coefficients: $2 = A + C = A$, so $A = 2$. Result: $2/x - 1/(x^2)$.
- 16** Use partial fractions to find the Maclaurin (power series) expansion of $1/((1-x)(2-x))$ up to the term in x^2 , first decomposing into the form $A/(1-x) + B/(2-x)$.
 $1 = A(2-x) + B(1-x)$. At $x = 1$: $1 = A(1)$, so $A = 1$. At $x = 2$: $1 = B(-1)$, so $B = -1$. Result:
 $1/(1-x) - 1/(2-x) = 1/(1-x) - 1/2 \cdot 1/(1-x/2)$. Expanding each as a geometric series:
 $(1 + x + x^2 + \dots) - 1/2(1 + x/2 + x^2/4 + \dots) = (1 - 1/2) + (1 - 1/4)x + (1 - 1/8)x^2 + \dots = 1/2 + 3/4x + 7/8x^2 + \dots$.
- 17** Express $(4x^2 - 5x - 15)/((x+1)(x-2)(x+3))$ in partial fractions, using the cover-up method (evaluating the numerator at each root while covering the matching factor in the denominator).
 Write $(4x^2 - 5x - 15)/((x+1)(x-2)(x+3)) = A/(x+1) + B/(x-2) + C/(x+3)$. Using the cover-up method: $A = (4(-1)^2 - 5(-1) - 15)/((-1-2)(-1+3)) = (4 + 5 - 15)/((-3)(2)) = -6/(-6) = 1$.
 $B = (4(2)^2 - 5(2) - 15)/((2+1)(2+3)) = (16 - 10 - 15)/((3)(5)) = -9/15 = -3/5$.
 $C = (4(-3)^2 - 5(-3) - 15)/((-3+1)(-3-2)) = (36 + 15 - 15)/((-2)(-5)) = 36/10 = 18/5$.
 Result: $1/(x+1) - 3/(5(x-2)) + 18/(5(x+3))$.

- 18** A chemical reaction's rate is modeled by $dC/dt = 1/((C + 2)(C - 1))$, where C is a concentration. Express $1/((C + 2)(C - 1))$ in partial fractions of the form $A/(C + 2) + B/(C - 1)$, and then use the result to write the general antiderivative $\int 1/((C + 2)(C - 1)) dC$ in terms of natural logarithms.
 $1 = A(C - 1) + B(C + 2)$. At $C = 1$: $1 = B(3)$, so $B = 1/3$. At $C = -2$: $1 = A(-3)$, so $A = -1/3$. Result: $-1/3/(C + 2) + 1/3/(C - 1)$. Integrating:
 $\int 1/((C + 2)(C - 1)) dC = -1/3 \ln|C + 2| + 1/3 \ln|C - 1| + K = 1/3 \ln|(C - 1)/(C + 2)| + K$.
- 19** Use the decomposition $1/(3x) + 8/(3(x - 3))$ found earlier in this worksheet for $(3x - 1)/(x(x - 3))$ to evaluate $\int_4^5 (3x - 1)/(x(x - 3)) dx$, giving your answer as a single logarithm expression.
 $\int (3x - 1)/(x(x - 3)) dx = 1/3 \ln|x| + 8/3 \ln|x - 3| + C$. Evaluating from 4 to 5:
 $[1/3 \ln 5 + 8/3 \ln 2] - [1/3 \ln 4 + 8/3 \ln 1] = 1/3 \ln 5 + 8/3 \ln 2 - 1/3 \ln 4 = 1/3 \ln(5/4) + 8/3 \ln 2$.
- 20** This question has three parts. (a) Factor $x^2 - 4$ and hence express $(x + 8)/(x^2 - 4)$ in partial fractions. (b) Use your result from part (a) to write down $\int (x + 8)/(x^2 - 4) dx$. (c) Hence evaluate $\int_3^5 (x + 8)/(x^2 - 4) dx$, giving your answer as a single expression involving natural logarithms.
(a) Factor $x^2 - 4 = (x - 2)(x + 2)$, so $(x + 8)/((x - 2)(x + 2)) = A/(x - 2) + B/(x + 2)$. Then $x + 8 = A(x + 2) + B(x - 2)$. At $x = 2$: $10 = 4A$, so $A = 5/2$. At $x = -2$: $6 = -4B$, so $B = -3/2$. Result: $5/2/(x - 2) - 3/2/(x + 2)$. (b) $\int (x + 8)/(x^2 - 4) dx = 5/2 \ln|x - 2| - 3/2 \ln|x + 2| + C$. (c) Evaluating from 3 to 5: $[5/2 \ln 3 - 3/2 \ln 7] - [5/2 \ln 1 - 3/2 \ln 5] = 5/2 \ln 3 - 3/2 \ln 7 + 3/2 \ln 5$ (since $\ln 1 = 0$).

Combining partial fractions back into a single fraction

- 21** Combine $2/(x + 1) + 3/(x + 2)$ into a single fraction.
The common denominator is $(x + 1)(x + 2)$. So
 $2/(x + 1) + 3/(x + 2) = (2(x + 2) + 3(x + 1))/((x + 1)(x + 2))$. Expanding the numerator:
 $2(x + 2) + 3(x + 1) = 2x + 4 + 3x + 3 = 5x + 7$. Result: $(5x + 7)/((x + 1)(x + 2))$ — confirming the original expression from the first question of this worksheet.

Identifying the correct form of a decomposition

- 22 Which of the following is the correct form for the partial fraction decomposition of $(3x + 1)/((x - 2)^2(x + 1))$?

a. $A/(x-2)+B/(x-2)^2+C/(x+1)$ ✓

b. $A/(x-2)+B/(x+1)$

c. $A/((x-2)^2)+B/(x+1)$

d. $(Ax+B)/(x-2)^2+C/(x+1)$

A repeated linear factor $(x - 2)^2$ requires two separate terms, $A/(x - 2)$ and $B/(x - 2)^2$, while the distinct linear factor $(x + 1)$ contributes one term, $C/(x + 1)$. The second option omits the repeated-factor structure, the third omits the lower-power term, and the fourth incorrectly uses a linear numerator over a repeated linear factor (that form is only needed for irreducible quadratic factors).

Solving a separable differential equation using partial fractions

- 23 Solve the differential equation $dP/dt = P(10 - P)/100$, given that $P = 5$ when $t = 0$. Use the partial fraction decomposition $1/(P(10 - P)) = 1/(10P) + 1/(10(10 - P))$ found earlier in this worksheet to integrate, and give P explicitly as a function of t .

Separating variables: $dP/(P(10 - P)) = dt/100$. Using the decomposition, the left side integrates to $(1/10)\ln|P| - (1/10)\ln|10 - P| = (1/10)\ln|P/(10 - P)|$. So $(1/10)\ln|P/(10 - P)| = t/100 + C_1$.

Multiplying by 10: $\ln|P/(10 - P)| = t/10 + C_2$. Exponentiating: $P/(10 - P) = ke^{t/10}$ for some constant k . At $t = 0$, $P = 5$, so $5/5 = 1 = k$. So $P/(10 - P) = e^{t/10}$, which rearranges to $P = 10e^{t/10}/(1 + e^{t/10})$.

Explaining when polynomial division is required

- 24 Explain why polynomial division must be performed before applying partial fraction decomposition to $x^3/(x - 1)^2$, but not to $(3x + 1)/(x - 1)^2$.

The standard partial fraction method requires the numerator's degree to be strictly less than the denominator's degree. In $x^3/(x - 1)^2$, the numerator has degree 3 while the denominator $(x - 1)^2$ has degree 2, so the fraction is improper and must first be reduced by polynomial division into a polynomial plus a proper fraction before decomposing. In $(3x + 1)/(x - 1)^2$, the numerator has degree 1, which is already less than the denominator's degree 2, so the fraction is already proper and can be decomposed directly as $A/(x - 1) + B/(x - 1)^2$.

Repeated irreducible quadratic factor

- 25** Express $(2x^3 + 3x^2 + 4x + 1)/(x^2 + 1)^2$ in partial fractions of the form $(Ax + B)/(x^2 + 1) + (Cx + D)/(x^2 + 1)^2$.
- $2x^3 + 3x^2 + 4x + 1 = (Ax + B)(x^2 + 1) + Cx + D$. Expanding: $(Ax + B)(x^2 + 1) = Ax^3 + Bx^2 + Ax + B$. Comparing coefficients: x^3 : $A = 2$. x^2 : $B = 3$. x : $A + C = 4$, so $C = 2$. Constant: $B + D = 1$, so $D = -2$. Result: $(2x + 3)/(x^2 + 1) + (2x - 2)/(x^2 + 1)^2$.
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Verifying a decomposition by substitution

- 26** Verify the decomposition $(5x + 7)/((x + 1)(x + 2)) = 2/(x + 1) + 3/(x + 2)$ found earlier in this worksheet by substituting $x = 0$ into both sides.
- LHS: $(5(0) + 7)/((0 + 1)(0 + 2)) = 7/2$. RHS: $2/(0 + 1) + 3/(0 + 2) = 2 + 3/2 = 7/2$. Since both sides equal $7/2$, the decomposition checks out at $x = 0$.
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Choosing efficient substitution values

- 27** When decomposing $(2x + 3)/((x - 4)(x + 5))$ using the equation $2x + 3 = A(x + 5) + B(x - 4)$, which two values of x should be substituted to find A and B most efficiently, and why?
- Substitute $x = 4$ and $x = -5$ — the roots of the denominator's factors — since each choice makes one of the two terms on the right vanish, isolating A or B directly without solving simultaneous equations. At $x = 4$: $11 = 9A$, so $A = 11/9$. At $x = -5$: $-7 = -9B$, so $B = 7/9$.